



AI Agents

Models that can choose tools, act, and adapt within a controlled workflow.

An agent independently works toward a task on your behalf.

The agent metaphor

It receives a mission, chooses among available capabilities, observes what happens, and continues until it finishes or must stop.



Three conditions make agentic workflows especially useful.

01

Complex decisions

The next step depends on context, evidence, or earlier results.

02

Rules are brittle

Maintaining every possible branch becomes difficult or expensive.

03

Unstructured data

The task depends on documents, conversations, images, or free-form text.

The quality of the system depends on all three working together.

01

MODEL

Reasons about the task and chooses the next step

02

TOOLS

Read data or take approved actions in external systems

03

INSTRUCTIONS

Define the mission, boundaries, priorities, and stopping rules

Different models offer different operating points.

Task complexity

How much reasoning, planning, or ambiguity must the model handle?

Latency

How quickly must the user receive a useful result?

Cost

How much inference spend is acceptable at the expected volume?

FIT

right fit
not
largest

Without tools, the agent can only reason and respond with text.

Q equips the agent.

Software tools provide the same leverage: access information, perform actions, or coordinate other workers.



Classify a tool by what changes after the call.

Read

DATA

Search the web, query a database, read a PDF, retrieve CRM context

Change

ACTION

Send email, update a record, create a ticket, write a file

Delegate

ORCHESTRATION

Call a specialist agent or route work through another workflow



A capable agent still needs an explicit objective, rules, and stopping conditions.

“Your mission, should you choose to accept it...”

Good instructions remove ambiguity before the agent begins acting.

Four practices convert vague intent into testable operating guidance.

✓ Use existing documents

Ground instructions in approved procedures and policies.

✓ Break down tasks

Ask the agent to work through explicit stages when complexity demands it.

✓ Define clear actions

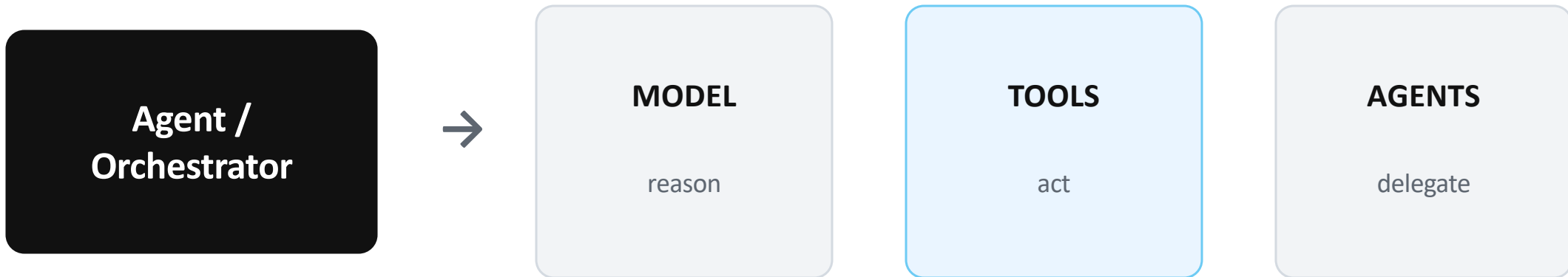
Specify when each tool may be used and what success looks like.

✓ Capture edge cases

State how to respond when data is missing, conflicting, or unsafe.

The conductor does not play every instrument; it controls timing and handoffs.

CONDUCTOR



Single-agent and multi-agent systems solve different coordination problems.

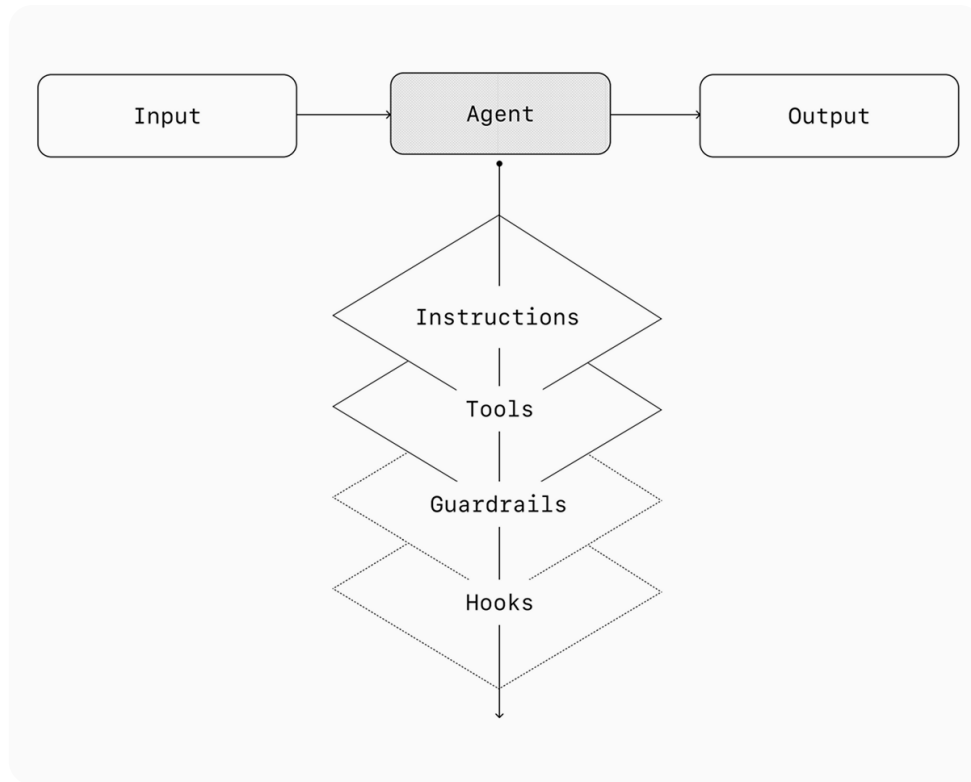


SINGLE AGENT



MULTI-AGENT SYSTEM

Add tools incrementally and keep the control loop observable.



- Add tools incrementally
- Keep complexity manageable
- Simplify evaluation and maintenance
- Define an exit condition
- Reuse prompt templates

Move beyond one agent only when the single-agent design becomes unreliable.

01

Complex logic

Distinct specialist roles make the workflow easier to reason about.

02

Tool overload

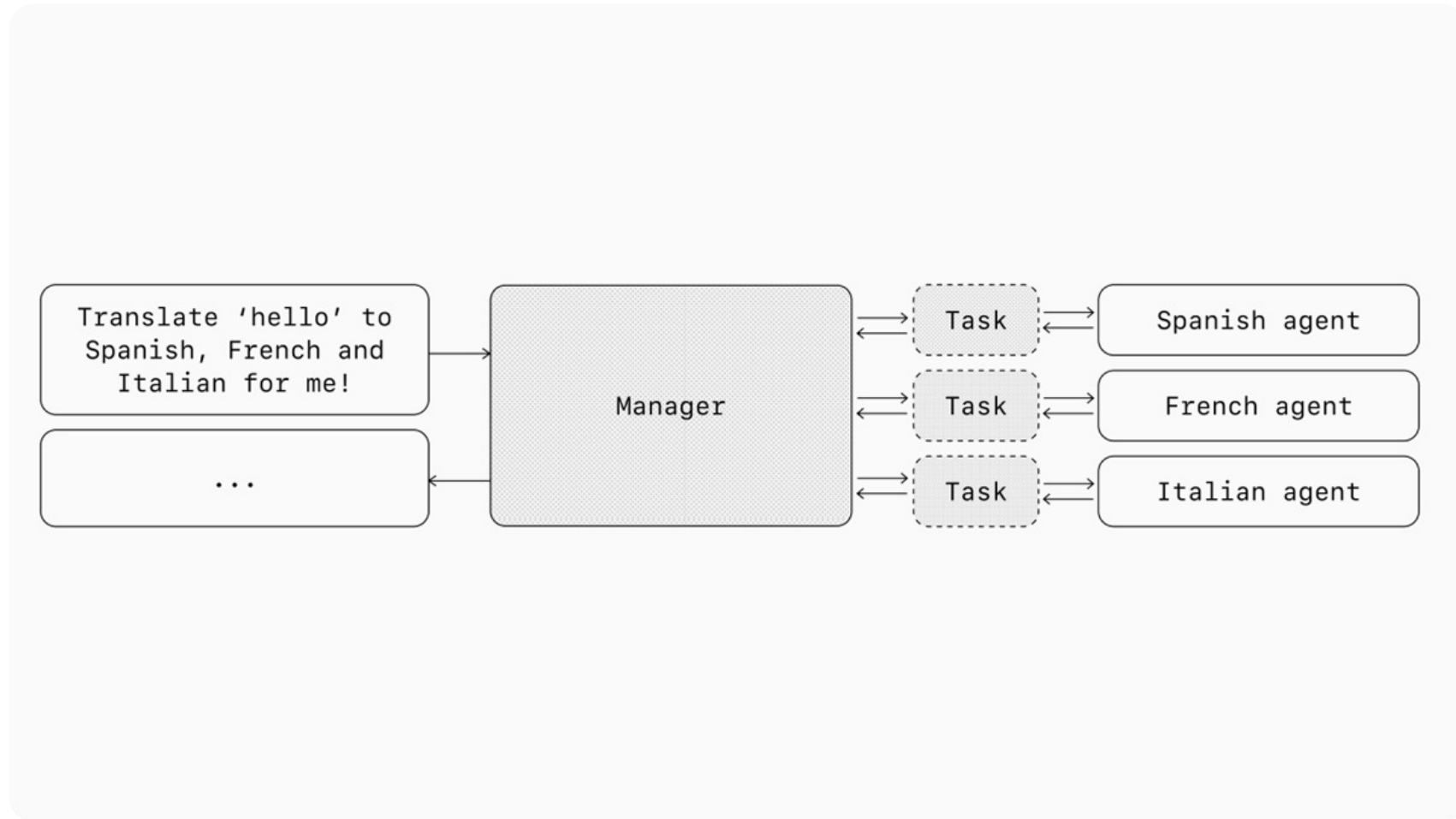
One agent cannot consistently select among too many overlapping tools.

03

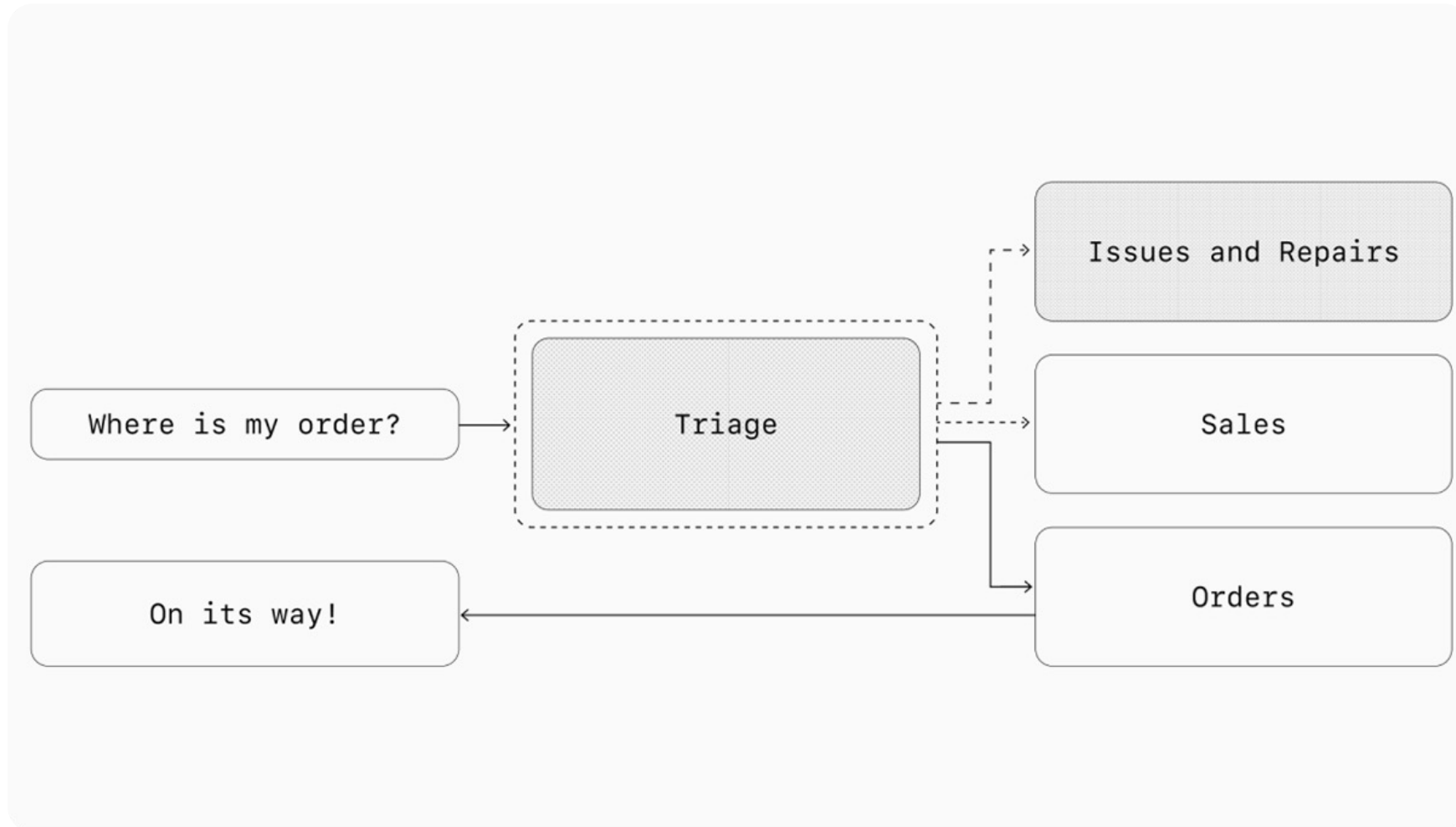
Instruction failure

The single agent repeatedly ignores required routing or procedure.

A central agent delegates to specialists and synthesizes the final result.



Specialists transfer the task when another agent becomes the better owner.



Adversarial examples reveal whether instructions and safeguards survive pressure.

Demo: Breaking Scammer AI



They detect risk, block unsafe behavior, validate outputs, or trigger escalation.

A guardrail is not the driver.

It reduces the impact of predictable failure modes while the model and workflow continue doing the task.



Choose controls according to the system's actual risks.

INPUT

- Relevance classifier
- Safety classifier
- PII filter

DURING ACTION

- Moderation
- Tool safeguards
- Rules-based protections

OUTPUT

- Output validation
- Policy and format checks
- Groundedness checks

Escalate when the workflow becomes repeatedly unreliable or the action carries significant risk.

FAILURE THRESHOLD

Repeated failures

Stop looping and send the history, evidence, and attempted actions to a person.

RISK BOUNDARY

High-risk actions

Require approval before consequential financial, legal, safety, or irreversible actions.

The safest agent knows when not to continue alone.

Build reliable agents deliberately, then improve them with evidence.

01

Strong Foundations

Pair a capable model with well-defined tools and clear, structured instructions.

02

Controlled Complexity

Start with one agent. Add orchestration only when needed, with guardrails and human oversight.

03

Iterative value

Validate with real users, then expand autonomy as the workflow proves reliable.
